Algorithm: A* search Algorithm

Source Code:

```cpp
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n, m;
    cin >> n >> m;
    vector<pair<int,int>> adj[100];
    for(int i=0,u,v,w;i<m;i++) {
        cin >> u >> v >> w;
        adj[u].push_back({v,w});
        adj[v].push_back({u,w});
    }
    vector<int> h(n), g(n, 1e9), parent(n, -1);
    for(int i=0;i<n;i++)
        cin >> h[i];
    int start, goal;
    cin >> start >> goal;
    priority_queue<pair<int,int>,
                   vector<pair<int,int>>,
                   greater<pair<int,int>>> pq;
    g[start] = 0;
    pq.push({h[start], start});
    while(!pq.empty()) {
        int u = pq.top().second;
        pq.pop();
        if(u == goal) break;
        for(auto x : adj[u]) {
            int v = x.first;
            int cost = x.second;
            if(g[u] + cost < g[v]) {
                g[v] = g[u] + cost;
                parent[v] = u;
                pq.push({g[v] + h[v], v});
            }
        }
    }
    cout << "Cost = " << g[goal] << endl;
    vector<int> path;
    for(int v = goal; v != -1; v = parent[v])
        path.push_back(v);
    reverse(path.begin(), path.end());
    cout << "Path: ";
    for(int x : path)
        cout << x << " ";
}
```

Output:

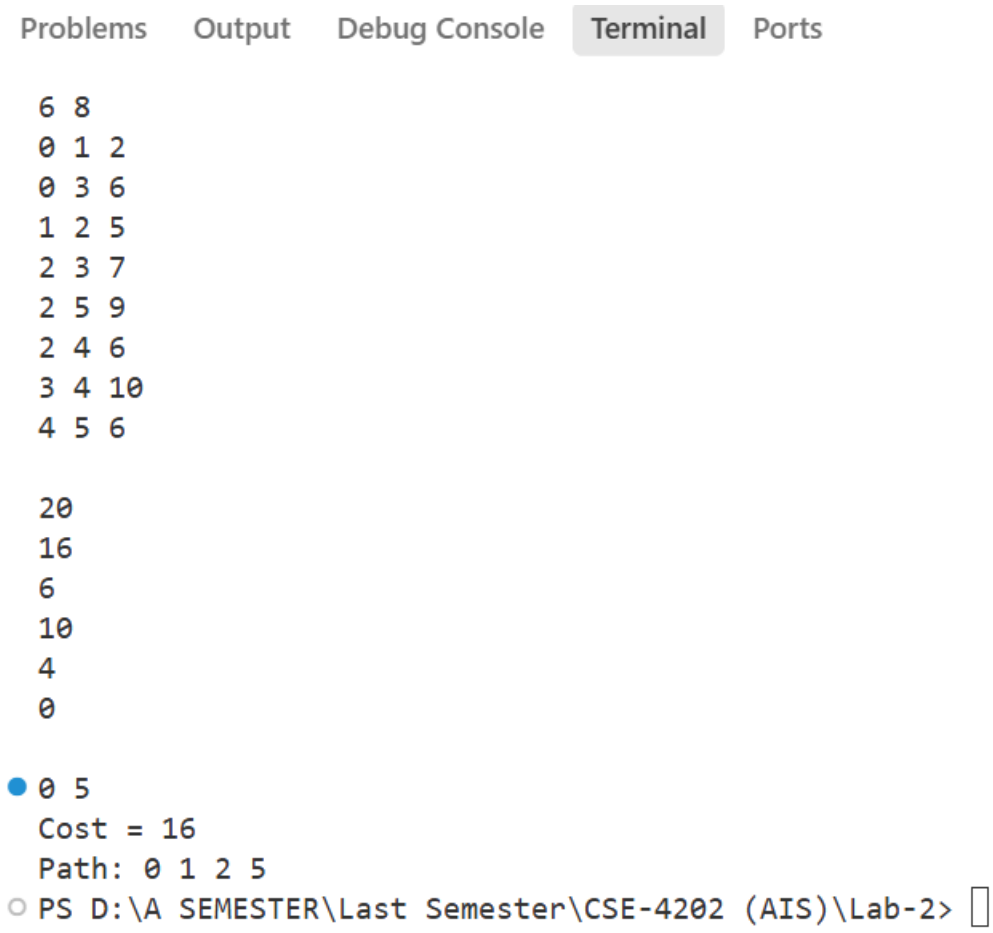


Fig: Output of A* search Algorithm